EV Charging Network

Performance & Business Analytics

An end-to-end analysis of 100,000 charging sessions across 8 Indian cities, 300 stations and 4 charger types (Jan 2024 - Dec 2025)

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What this deck covers

Project overview and roadmap

Question: how is a nationwide EV charging network really performing - where does revenue come from, who waits, who is happy?

Tools used: Python (pandas, NumPy, Matplotlib, Seaborn) for cleaning and EDA; Power BI for the two-page interactive dashboard (Business Overview and Operations & Customer Experience).

1. Data and cleaning (slides 3-4)
2. Headline KPIs (5)
3. When people charge (6)
4. Where: cities (7)
5. What: chargers, vehicles, stations, payments (8-11)
6. Experience: wait time and ratings (12)
7. Seasonality (13)
8. Insights and recommendations (14-15)

The dataset

One row = one charging session

Session_ID / Session_Start	Unique ID and timestamp
City / Station_ID / Station_Type	8 cities, 300 stations, 5 types
Charger_Type	AC 7kW, DC 30kW, DC 60kW, DC 120kW
Vehicle_Type	Bike, Car, SUV, Taxi, Bus
Energy_kWh	Energy delivered (8 to 95 kWh)
Wait_Time_Min	Queue time (0 to 219 min)
Charging_Duration_Min	10 to 120 min
Revenue_INR	Rs 80 to Rs 2,082 per session
Customer_Rating	1 to 5 stars
Payment_Mode / Hour / Day / Month	Card, UPI, Wallet, Fleet; time features

Data preparation

Cleaning steps from the Python notebook

- Inspected shape, data types, summary statistics and unique counts for every column.
- Checked missing values and removed duplicate rows so no session is counted twice.
- Filled missing City and Charger_Type with the most common value (mode); filled missing Wait_Time, Revenue and Rating with the median, which is robust to outliers.
- Reviewed outliers in energy, wait time, duration, revenue and rating using box plots and IQR logic.
- Created Hour, Day and Month from Session_Start, then exported the clean file (EV_Cleaned.csv) for Power BI.

Result

100,000

clean sessions

16

columns, 0 nulls

Headline KPIs

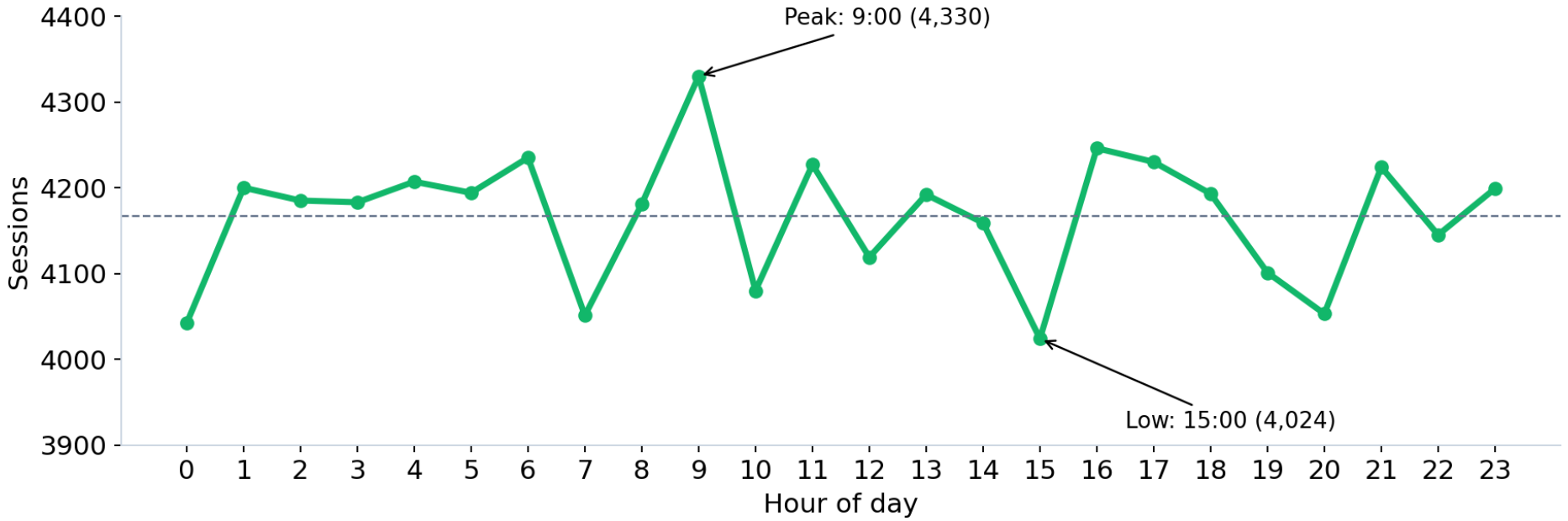
Matches the Power BI Business Overview page

100K	Total sessions
Rs 82.3M	Total revenue
5.15M kWh	Total energy
23.3 min	Avg wait time
3.00 / 5	Avg rating
64.9 min	Avg charge duration
51.5 kWh	Avg energy / session
Rs 823	Avg revenue / session

Pricing works out to roughly Rs 16 per kWh on average. The network is large and steady, but a 3.0 average rating and a 23-minute average wait leave clear room to improve.

When do people charge?

Sessions by hour of day

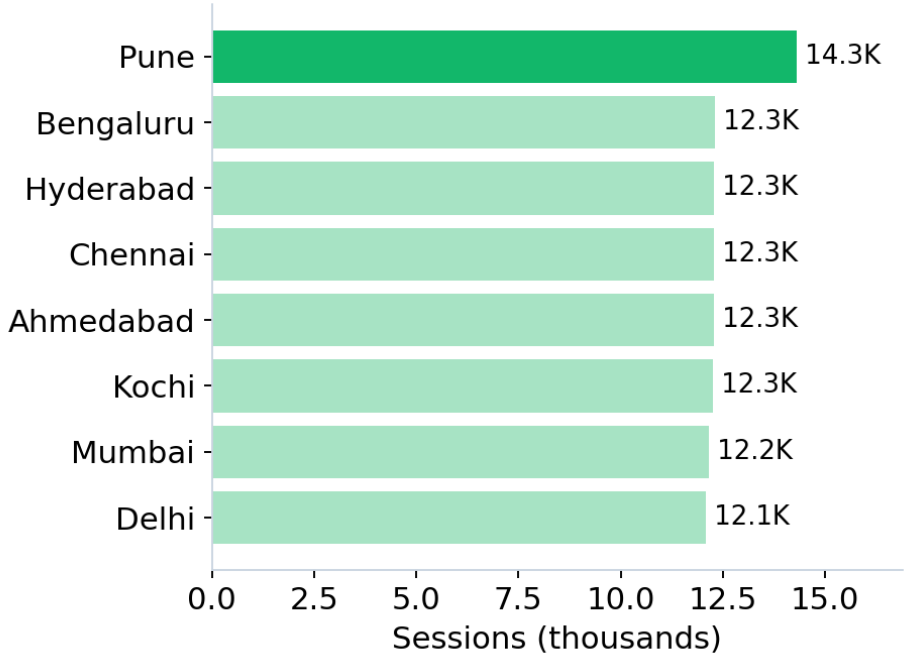


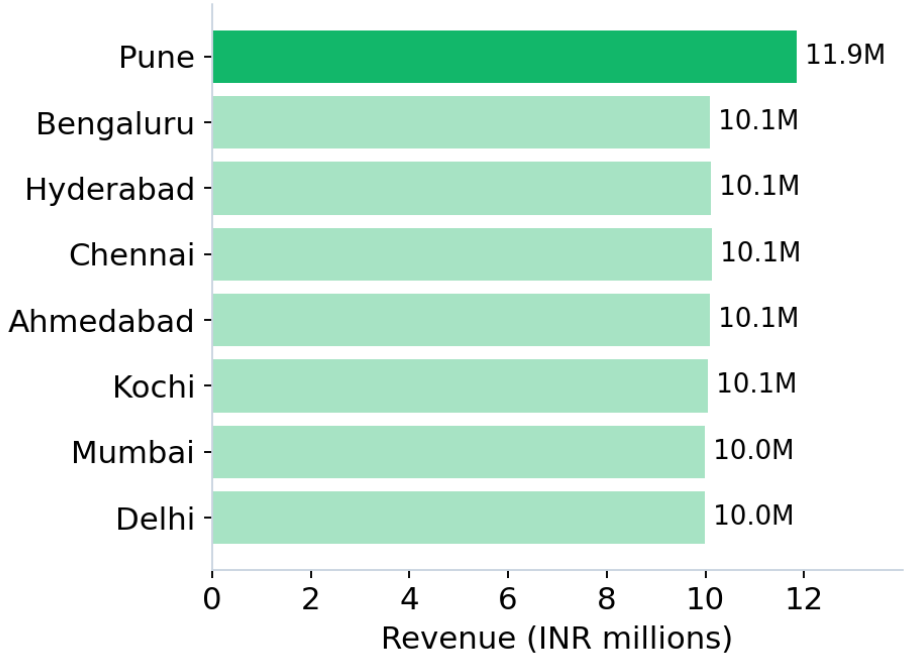
Demand is remarkably flat around the clock: every hour sees between 4,024 and 4,330 sessions (about 4,170 on average). The busiest hour (9:00) is only ~8% above the quietest (15:00).

So what: there is no sharp rush hour, which means capacity can be planned evenly and time-of-day pricing will have limited effect unless the pattern shifts.

Where: sessions and revenue by city

Pune leads; the other seven cities are almost level



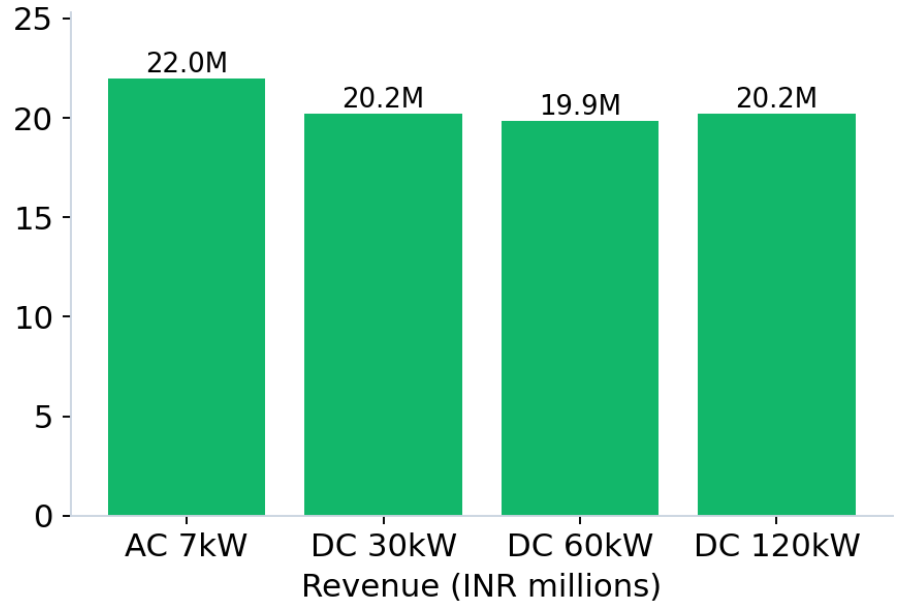


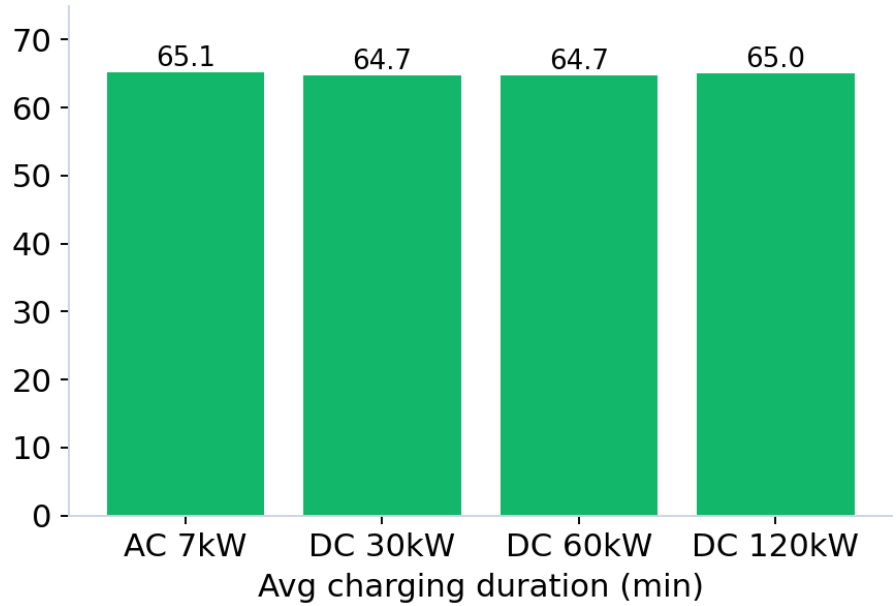
Pune handles 14.3K sessions and Rs 11.9M revenue, about 16-17% more than any other city. The remaining seven sit within 12.1K-12.3K sessions and Rs 10.0M-10.1M each.

So what: Pune is the proven flagship market. Study what is different there (station mix, density, location) before expanding elsewhere.

What: charger types

AC 7kW is used most; all types earn similar revenue per session



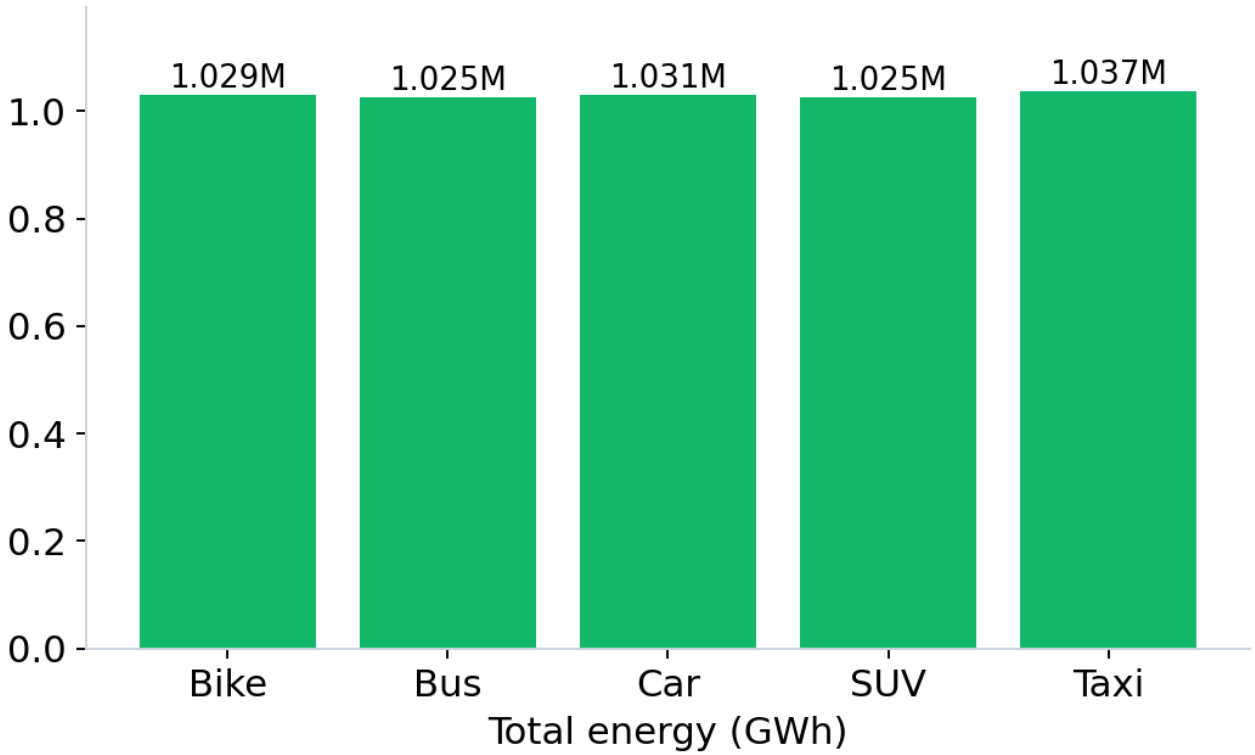


AC 7kW has the most sessions (26.7K) and revenue (Rs 22.0M). DC 30/60/120 kW each carry roughly 24.3K-24.5K sessions. Revenue per session is Rs 818-826 for all four.

Odd finding: a 120 kW fast charger has about the same average charging duration (~65 min) as a 7 kW one. In real networks fast chargers finish much sooner, so this is worth validating at source.

What: vehicle types

Energy delivered by vehicle category

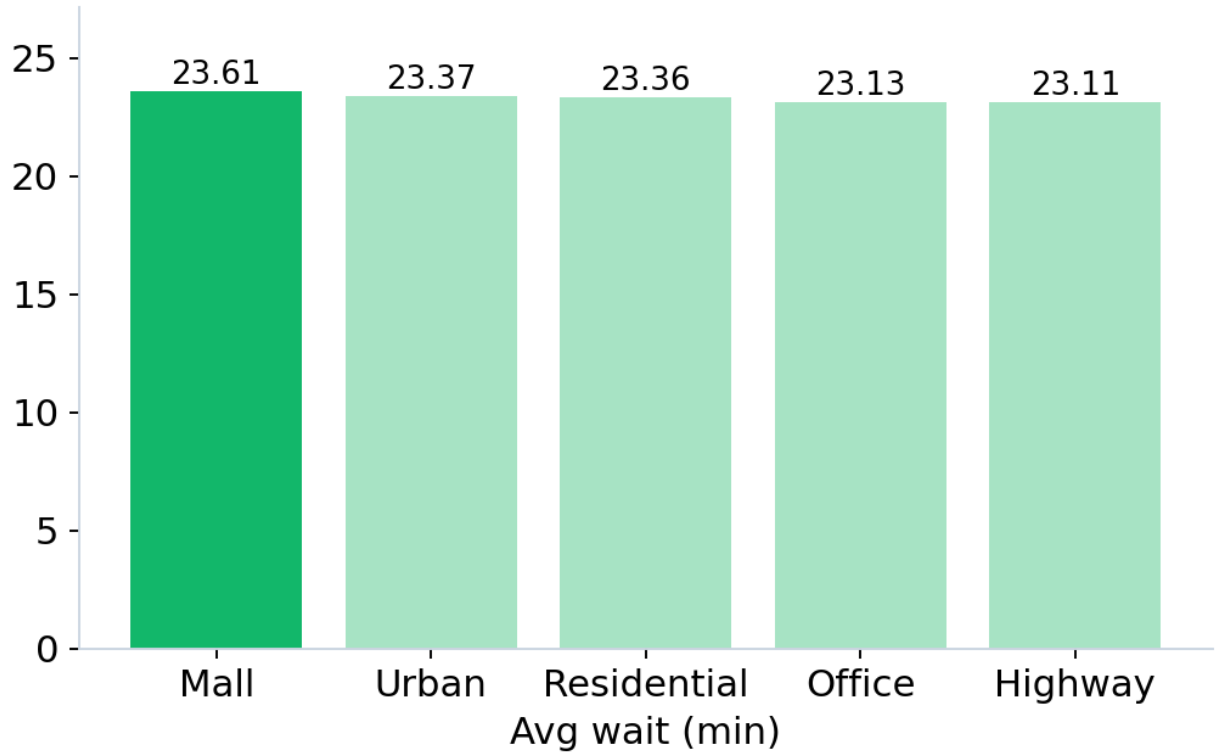


- Each type draws about 1.03M kWh.
- Taxi leads slightly (20.2K sessions), but the gap to Bike, Car, Bus and SUV is under 1%.
- A bus and a bike use the same energy per session (~51 kWh), which is unusual in the real world.

So what: the customer base is well diversified, with no dependence on one vehicle segment.

What: station types

Average wait time by location type

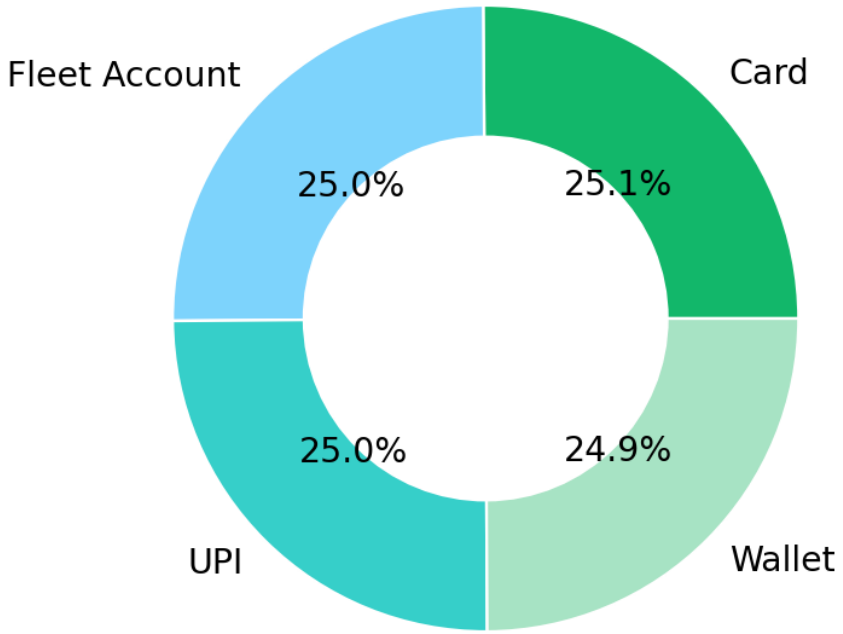


- Highway stations see the most sessions (20.3K).
- Mall stations have the longest average wait (23.6 min); Highway and Office are shortest (23.1 min).
- Urban stations earn the most per session (Rs 829); Mall the least (Rs 819).

So what: the differences are small (under 0.5 min), so wait time is a network-wide issue, not a location problem.

Payment behaviour

Share of sessions by payment mode

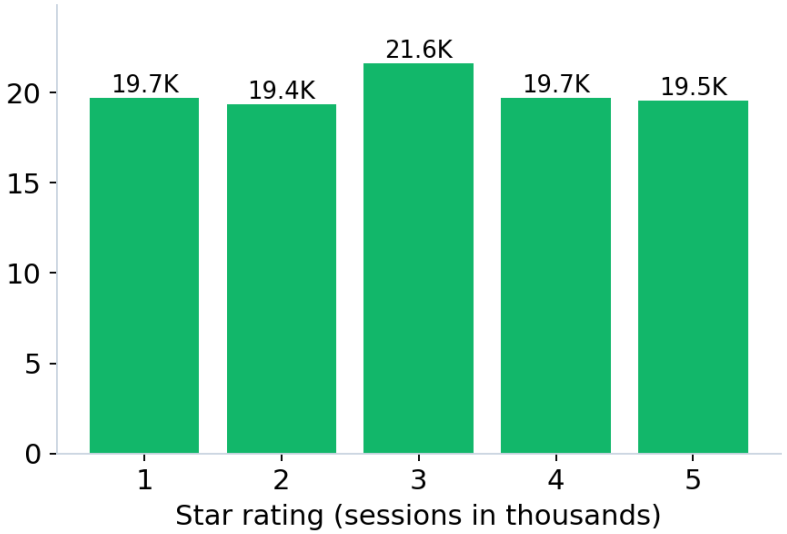


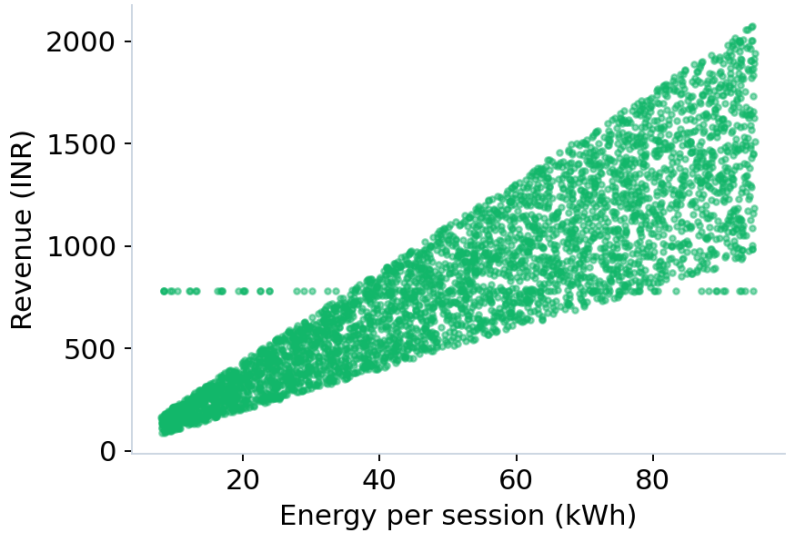
- Card, UPI, Wallet and Fleet Account each hold about 25% of sessions.
- Fleet Account customers pay the most per session (Rs 828) and generate the highest total revenue (Rs 20.7M).
- Wallet is lowest per session (Rs 820), but the spread is only Rs 7.

So what: digital payments are fully mainstream. A fleet account is a natural route to B2B contracts with taxi and bus operators.

Customer experience: wait time and ratings

Operations & Customer Experience page



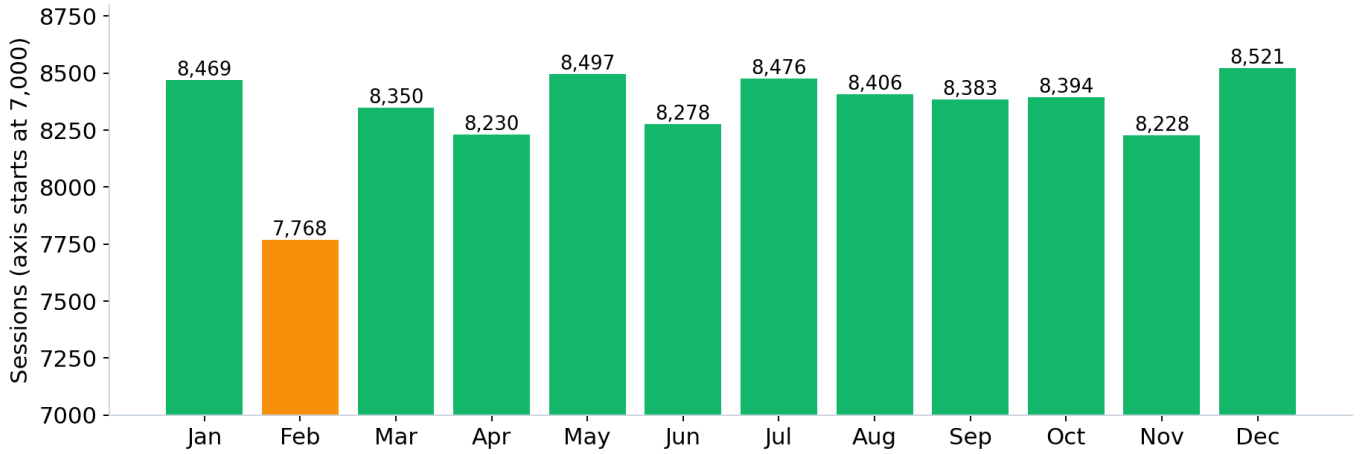


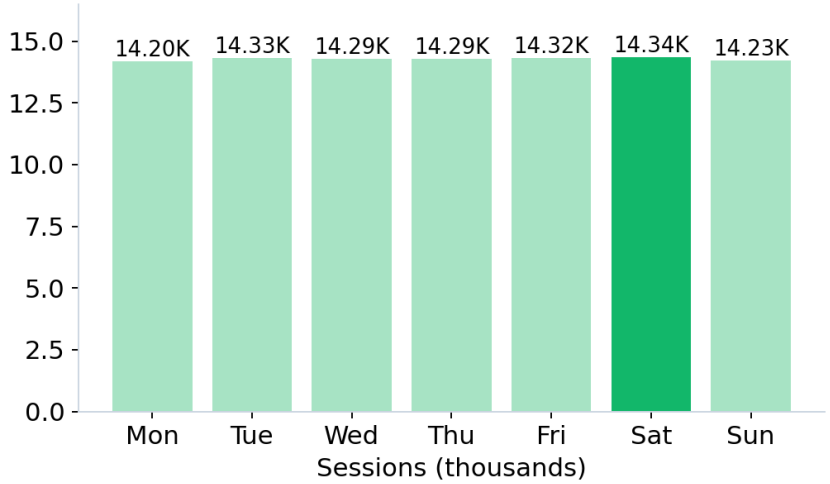
Ratings are spread almost evenly from 1 to 5 stars (about 19.4K-21.6K each) and average 3.00. Wait time shows zero correlation with rating (r = 0.00), and revenue tracks energy closely (r = 0.89).

Important caveat: in real data, longer waits normally lower ratings. A flat result like this suggests the ratings may be randomly generated or not linked to the session. Before acting, confirm how ratings are collected.

Seasonality: month and weekday

Sessions per month (top) and by weekday





- February is the lowest month (7,768 sessions), mainly because it has fewer days.
- July, December, January and May are the strongest months.
- Weekdays are level: Saturday is highest (14.3K), Monday lowest (14.2K).

Key insights

1 Balanced network

Cities, chargers, vehicles, stations and payment modes are all within a few percent of each other.

2 Pune stands out

The one clear outlier: about 16% more sessions and revenue than any other city.

3 Revenue follows energy

Revenue per session is Rs 823 and depends almost entirely on kWh delivered (r = 0.89).

4 Experience is flat

Average wait is 23 minutes and ratings average 3.0, with no link between the two.

Recommendations and conclusion

- Replicate Pune: audit its station layout, locations and utilisation, then apply the lessons in the other seven cities.
- Cut wait time: add chargers or booking where the 23-minute average and the long tail (up to 219 min) hurt most.
- Fix the rating signal: link ratings to the session and add short reason tags so feedback becomes actionable.
- Validate charger data: confirm that DC 120 kW sessions really last as long as AC 7 kW ones.
- Grow fleet business: Fleet Account is already the top revenue payment mode, so offer taxi and bus operators contracts.

Conclusion: a healthy, evenly spread EV network with Rs 82.3M revenue. The next gains come from fixing waiting time and customer feedback quality rather than from any single city or charger type.